

# Random Variables

$\Omega$ : set of possible outcomes.

↑  
numerical.

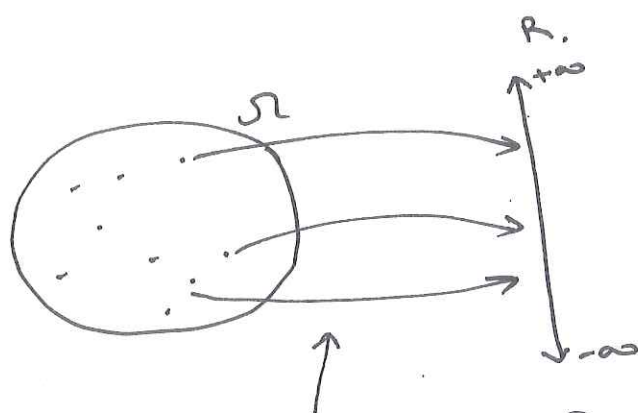
ex: balls numbered 1, 2, ... 50  
note # of ball.

not numerical.

ex: toss a coin 3 times  
note the seqn.

↑  
assign a number  
to each outcome.  
how?

ex: assign a number  
to each seqn.  
note # of  
heads in the  
seqn.



rule assigning  $\Omega$  to  $R$ : random variable.  
real-valued fn. of the experimental outcome.

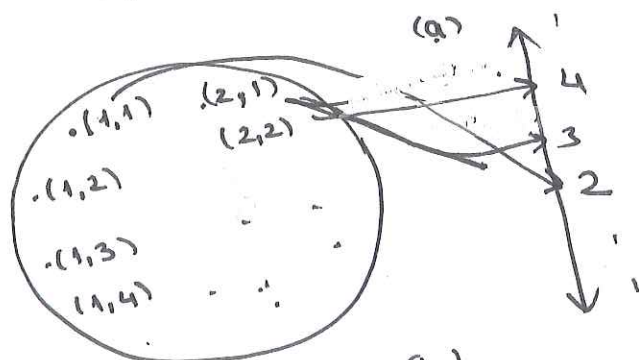
ex: experiment involving two rolls of a die.

example random variables

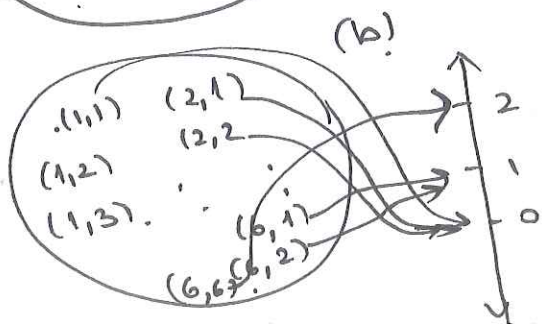
a)  $X$  = sum of two rolls

b)  $Y$  = # of sixes in two rolls

c) second roll raised to fifth power



$\Omega = \{(1,1), (1,2), \dots\}$   
 $\Omega_X = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$



$\Omega = \{(1,1), (1,2), \dots\}$   
 $\Omega_Y = \{0, 1, 2\}$

Formal Definition: A random variable  $X$  is a fn. st. (2)

$$\forall \omega \in \Omega : X(\omega) \in \mathbb{R}$$

$$\text{or } X: \Omega \rightarrow \mathbb{R}_X \subset \mathbb{R}.$$

Why need random variable?

→ easier mathematical manipulation  
 → associate with each r.v. certain 'averages' of interest such as mean and variance.

→ fn. of a r.v. is another r.v.

→ r.v. conditioned on an event is another r.v.

Types of random variables:

a) Discrete r.v: r.v. whose range is either finite or countably infinite

ex: 1) # of heads in 3 coin tosses:  $\{0, 1, 2, 3\}$

2) # of defective items chosen out of 100 items containing 5 defective ones. (choose 10 items):  $\{0, 1, 2, 3, 4, 5\}$

b) Continuous r.v: r.v. that can take on uncountably infinite # of values.

ex: choose a point  $a$  from interval  $[-1, 1]$  that associates  $a^2$  to outcome  $a$ .

random variable  $f(a) = a^2$  → continuous.

random variable  $g(a) = \sin(a)$  that associates  $\sin(a)$  to outcome  $a$ .

Discrete r.v:

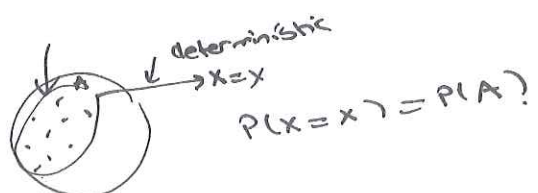
How to characterize a r.v.?  
 → probabilities of the values that it can take.

Probability mass function (pmf).

$$P_X(x) = P(\{X=x\}) \text{ for every number } x \text{ s.t.}$$

$$1) \forall x \in \mathbb{R}, P_X(x) \geq 0$$

$$2) \sum_{x \in \mathbb{R}} P_X(x) = 1.$$



ex: Experiment of tossing a fair coin 3 times.  
Note the # of heads.

(3)

$$\Omega = \{ HHH, HHT, HTH, HTT, THH, THT, TTH, TTT \}$$

$$X(\Omega) = X = \left\{ \begin{array}{cccccccc} \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 3 & 2 & 2 & 1 & 2 & 1 & 1 & 0 \end{array} \right\}$$

$$\Omega_X = \{ 0, 1, 2, 3 \}$$

$$P_X(0) = P(X=0) = P(\{ TTT \}) = \frac{1}{8}$$

$$P_X(1) = P(X=1) = P(\{ HTT, THT, TTH \}) = \frac{3}{8}$$

$$P_X(2) = P(X=2) = P(\{ HHT, HTH, THH \}) = \frac{3}{8}$$

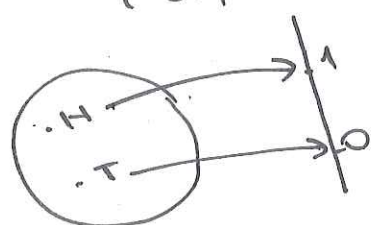
$$P_X(3) = P(X=3) = P(\{ HHH \}) = \frac{1}{8}$$

Specific r.v.s:

Bernoulli r.v.:

A r.v. whose only possible values are 0 and 1.  
ex: observing heads or tails from the toss of a fair coin

$$X = \begin{cases} 1, & \text{heads} \\ 0, & \text{tails} \end{cases}$$

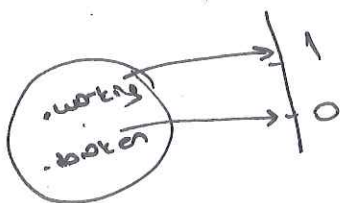


$$P_X(0) = P(X=0) = \frac{1}{2}$$

$$P_X(1) = P(X=1) = \frac{1}{2}$$

ex: A machine  $\rightarrow$  broken or working

$$X = \begin{cases} 1, & \text{working} \\ 0, & \text{broken} \end{cases}$$



$$P_X(0) = P(X=0) = 0.1$$

$$P_X(1) = P(X=1) = 0.9$$

Bernoulli r.v: pmf:

$$P_X(x) = \begin{cases} 1-p, & x=0 \\ p, & x=1 \\ 0, & \text{otherwise} \end{cases}$$

Binomial r.v:

A coin is tossed  $n$  times.  
At each toss, the coin comes up head with prob.  $p$ ,  
and tail up. " $(1-p)$ "

$X$ : # of heads in  $n$ -toss seqn.

$X$ : binomial r.v. with parameters  $n$  and  $p$ .

$$P_X(k) = P(X=k) = \binom{n}{k} p^k (1-p)^{n-k} \quad k=0, 1, \dots, n.$$

$$\sum_{k=0}^n \binom{n}{k} p^k (1-p)^{n-k} = 1.$$

An experiment is called a binomial exp. if

- 1) experiment consists of a seqn. of  $n$  trials, where  $n$  is fixed
- 2) trials are identical and indep. with two possible outcomes, S or F, where prob of success is  $p$ .

ex: In a nuclear reactor, the fission is controlled by inserting into the radioactive core a number of special rods. (that absorb neutrons and slow down the nuclear chain reaction) Assume that a particular reactor has 10 of these rods, each operating independently and each having 0.8 probability of functioning properly. A meltdown is prevented if at least half of the rods perform satisfactorily. what is the probability that system will fail?

$X$ : # of rods that perform satisfactorily.  
 $n: 10, \quad p=0.8$

$$P(\text{system fail}) = P(X < 5) = \sum_{k=0}^4 \binom{10}{k} (0.8)^k (0.2)^{10-k}$$

## Geometric r.v.:

Sequential and independent Bernoulli trials with individual probability of success  $p$ .

$X$ : total # of attempts until a success.

$X \rightarrow$  geometric r.v.

$$P_X(x) = P(X=x) = \underbrace{(1-p)^{x-1}}_{\substack{\uparrow \\ k-1 \text{ failures}}} \underbrace{p}_{\substack{\uparrow \\ 1 \text{ success}}} \quad x=1, 2, 3, \dots$$

$$\sum_{k=1}^{\infty} P_X(k) = \sum_{k=1}^{\infty} (1-p)^{k-1} p = p \cdot \sum_{k=0}^{\infty} (1-p)^k = p \cdot \frac{1}{1-(1-p)} = 1$$

ex:  $X$ : # of times a fair die is rolled until a '6' occurs.

$X$ : geometric r.v. with  $p = \frac{1}{6}$

$$P_X(1) = \frac{1}{6}$$

$$P_X(2) = \frac{5}{6} \cdot \frac{1}{6}$$

$$P_X(3) = \left(\frac{5}{6}\right)^2 \cdot \frac{1}{6}$$

$$P_X(k) = \left(\frac{5}{6}\right)^{k-1} \cdot \frac{1}{6}$$

## Poisson r.v.:

A r.v.  $X$  is said to have a Poisson distr. if

pmf of  $X$  is

$$P_X(x) = \frac{e^{-\lambda} \lambda^x}{x!} \quad x=0, 1, \dots$$

for some  $\lambda > 0$ , rate per unit time or per unit area

$$\sum_{k=0}^{\infty} \frac{e^{-\lambda} \lambda^k}{k!} = e^{-\lambda} \cdot \left(1 + \lambda + \frac{\lambda^2}{2!} + \frac{\lambda^3}{3!} + \dots\right) = e^{-\lambda} \cdot e^{\lambda} = 1$$

Application of Poisson r.v.:

events occur randomly in time or space.

Poisson r.v. to count # of occurrences of an event in a certain time period or in a certain region in space.

One other application of Poisson r.v.: approximate binomial probability.

Proposition: Binomial pmf  $\text{bin}(x; n, p)$ ,  
 $n \rightarrow \infty, p \rightarrow 0$  s.t.  $np \rightarrow \lambda > 0$ .

Then  $\text{bin}(x; n, p) \rightarrow \text{poi}(x; \lambda)$

Proof: need to show that for  $n$  large,  $p$  small,  $\alpha = np$ .  

$$P_k = \binom{n}{k} p^k (1-p)^{n-k} \sim \frac{\alpha^k \cdot e^{-\alpha}}{k!} \quad k=0, 1, \dots$$

First, consider the probability that no events occur in  $n$  trials

$$P_0 = \binom{n}{0} p^0 (1-p)^n = (1-p)^n = \left(1 - \frac{\alpha}{n}\right)^n \rightarrow e^{-\alpha} \text{ as } n \rightarrow \infty \text{ (from calculus!)}$$

For the rest of the probabilities:

$$\frac{P_{k+1}}{P_k} = \frac{\binom{n}{k+1} p^{k+1} q^{n-k-1}}{\binom{n}{k} p^k q^{n-k}} = \frac{\frac{n!}{(k+1)!(n-k-1)!} p^{k+1} q^{n-k-1}}{\frac{n!}{k!(n-k)!} p^k q^{n-k}} = \frac{n!}{(n-k-1)!(k+1)!} \cdot \frac{p^{k+1} q^{n-k-1}}{p^k q^{n-k}}$$

$$= \frac{(n-k)}{(k+1)!} \cdot \frac{p}{q(1-p)} = \frac{\alpha \cdot (1-k/n)}{(k+1)(1-\alpha/n)} \rightarrow \frac{\alpha}{k+1} \text{ as } n \rightarrow \infty$$

so in the limit  $P_{k+1} = \frac{\alpha}{k+1} P_k$  for  $k=0, 1, 2, \dots$   
 and  $P_0 = e^{-\alpha}$ .

$$P_1 = \frac{\alpha}{1} \cdot P_0 = \frac{\alpha}{1} e^{-\alpha}$$

$$P_2 = \frac{\alpha}{2} \cdot P_1 = \frac{\alpha^2}{2 \cdot 1} e^{-\alpha}$$

$$\vdots$$

$$P_k = \frac{\alpha^k}{k!} e^{-\alpha}$$

Rule of thumb:  $n \geq 100, p \leq 0.01, np \leq 20 \Rightarrow$  approximate with Poisson

ex: The probability of a bit error in a comm. line is  $10^{-3}$ . find the probability that a block of 1000 bits has five or more errors.

Each bit transmission: Bernoulli with 'success' corresponding to a bit error.

Probability of  $k$  error in 1000 transmission:  $n = 1000$  and  $p = 10^{-3}$ .  $1000 \cdot 10^{-3} = 1$ .

Probability of  $x$  successes in  $n$  trials with  $p = 10^{-3}$  and  $n = 1000$  is given by Binomial distribution.

→ Poisson approximation:  $\alpha = np = 1000 \cdot 10^{-3} = 1$

$$P(X=x) = \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x} = \frac{n!}{x!(n-x)!} \frac{\alpha^x}{n^x} \left(1 - \frac{\alpha}{n}\right)^{n-x}$$

Binomial with  $\alpha = np = 1$   
 Poisson approximation:  
 $P(N \geq 5) = 1 - P(N < 5) = 1 - \sum_{k=0}^4 \frac{\alpha^k}{k!} e^{-\alpha}$   
 $= 1 - e^{-1} \cdot \left( 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} \right)$   
 $= 0.00366$

→ Exact expression:

→ Exact expression:

$$P(N \geq 5) = 1 - P(N < 5) = 1 - \sum_{k=0}^4 \frac{(1000)^k}{k!} \cdot (10^{-3})^k \cdot (1 - 10^{-3})^{1000-k}$$

↑  
hard to calculate.

At a

ex: An ISP has  $c$  nodes and  $n$  customers. At a given time, each customer needs a connection with prob.  $p$ .

Now:  $c=50$ ,  $n=1000$ ,  $p=0.01$ .

Now:  $c=50$ ,  $n=1000$ ,  $p=0.001$

a) pmf of # of modems in use? : X

$\{0\}$  use  $\} = \{1$  customer needs a connection  $\}$

$\{1$  modem in use  $\} = \{2$  " " " "  $\}$

$\{2$  " " " " " " " " " "  $\}$

$$P(X=0) = \binom{n}{0} p^0 (1-p)^n$$

$$P(X=1) = \binom{n}{1} p^1 (1-p)^{n-1}$$

$$P(X=2) = \binom{n}{2} p^2 (1-p)^{n-2}$$

$$P(X=50) = \sum_{k=0}^n \binom{n}{k} p^k (1-p)^{n-k}$$

b) approx. pmf of  $\hat{\alpha}$ . (d = np = 10.)

approx. pmf of  $\text{mean} =$   
 $P(X = i) = \frac{e^{-\alpha} \alpha^i}{i!}$   
 $P(X = i) = \frac{\alpha^i e^{-\alpha}}{i!}$   
 $i \in [0, 49]$   
 $\alpha = np = 10.$

$$P(X=50) = \frac{\sum_{k=0}^n \frac{x^k \cdot e^{-x}}{k!}}{ii.} =$$

c) probability that there are more customers than there are nodes? (8)

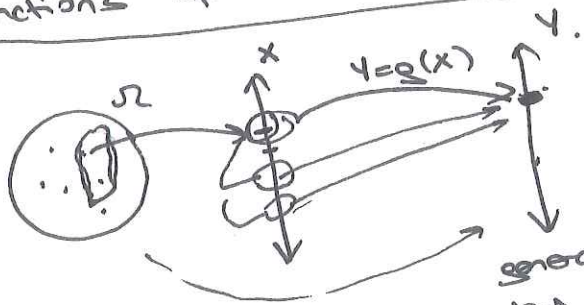
$P(\geq 51 \text{ customers need connection})$

$$= \sum_{k=c+1}^n \binom{n}{k} p^k (1-p)^{n-k}$$

d). approx. prob. that there are more customers than there are nodes?

$$P(\geq 51 \text{ customers}) = \sum_{k=c+1}^n e^{-\alpha} \frac{\alpha^k}{k!}$$

Functions of Random variables:



generate other r.v. by applying transformations on  $X$ .

$X$ : today's temperature in degrees celsius  
 $Y = 1.8X + 32$ : " " " " Fahrenheit.

$X$  r.v.  $\rightarrow Y = g(X)$  r.v. since numerical value for each possible outcome.

know pmf  $P_X$ , how to find pmf  $P_Y$ ?

$$P_Y(y) = \sum_{\{x | g(x)=y\}} P_X(x)$$

ex:  $Y = |X|$ ,  $P_X(x) = \begin{cases} 1/9, & \text{if } x \text{ integer in } [-4, 4] \\ 0, & \text{otherwise} \end{cases}$

$P_Y(y) = ?$

possible values for  $Y$ : 0, 1, 2, 3, 4.

$$P_Y(0) = \sum_{\{x | |x|=0\}} P_X(x) = P_X(0) = \frac{1}{9}$$

$$P_Y(1) = \sum_{\{x | |x|=1\}} P_X(x) = P_X(1) + P_X(-1) = \frac{2}{9}$$

$$P_Y(2) = P_X(2) + P_X(-2) = \frac{2}{9}$$

$$P_Y(3) = \frac{2}{9}$$

$$P_Y(4) = \frac{2}{9}$$

## Expectation:

pmf  $\rightarrow$  detailed info about a r.v.

expectation  $\rightarrow$  avg. of all possible values for a r.v.

ex: Tossing 3 coins

$X$ : # of H's in 3 coin tosses.

repeat experiment 20 times with the following outcome.

| # of heads:      | 0 | 1 | 2 | 3 |
|------------------|---|---|---|---|
| # of experiment: | 3 | 7 | 8 | 2 |

what is the average number of H's in an experiment?

$$\frac{0 \times 3 + 1 \times 7 + 2 \times 8 + 3 \times 2}{20} = 1.45$$

$$= 0 \times \left(\frac{3}{20}\right) + 1 \times \left(\frac{7}{20}\right) + 2 \times \left(\frac{8}{20}\right) + 3 \times \left(\frac{2}{20}\right)$$

frequency of observing 0 in 20 experiment.

Average of the r.v.;  $X = ?$

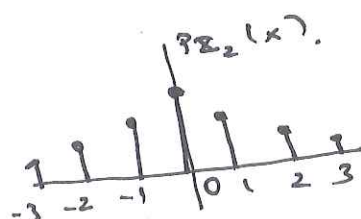
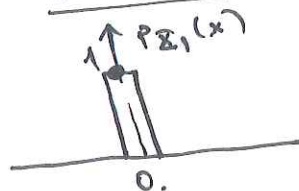
$$0 \times P_X(0) + 1 \times P_X(1) + 2 \times P_X(2) + 3 \times P_X(3)$$

$$= 0 \cdot \frac{1}{8} + 1 \cdot \left(\frac{3}{8}\right) + 2 \cdot \left(\frac{3}{8}\right) + 3 \cdot \left(\frac{1}{8}\right) = 1.5$$

Definition: The expected value (expectation or mean) of r.v.  $X$  with pmf  $P_X(x)$  is

$$E[X] = \sum x \times P_X(x)$$

## Variance:



both  $X_1$  and  $X_2$  mean = 0.

but variation around mean diff.  
Variance shows the degree of spread around mean.

Variance:

$$\text{var } X = E[(X - E(X))^2]$$

(10)

Standard deviation:

$$\sigma_X = \sqrt{\text{var}(X)}$$

ex: Let  $X$  have distrn.

$$P_X(x) = \begin{cases} 1/9, & \text{if } x \text{ integer in } [-4, 4] \\ 0, & \text{otherwise} \end{cases}$$

$$E[X] = \sum_x x \cdot P_X(x)$$

$$= \sum_{x=-4}^4 x \cdot \frac{1}{9} = \frac{1}{9} \sum_{x=-4}^4 x = 0$$

$$\text{var}(X) = E[(X - E(X))^2] = E[X^2] = E[Z] \text{ where } Z = X^2$$

$$P_Z(z) = \begin{cases} 1/9, & z=0 \\ 1/9, & z=1, 4, 9, 16 \\ 0, & \text{otherwise} \end{cases}$$

$$\text{var}(X) = E[Z] = \sum_z z \cdot P_Z(z) = 0 \cdot \frac{1}{9} + 1 \cdot \frac{2}{9} + 4 \cdot \frac{2}{9} + 9 \cdot \frac{2}{9} + 16 \cdot \frac{2}{9} = \frac{60}{9}$$

Here find distrn of  $Z = (X - E(X))^2 \rightarrow E[Z]$ .  
Easier method:

$$E[g(X)] = \sum_x g(x) \cdot P_X(x)$$

so,

$$\text{var}(X) = E[(X - E(X))^2] = \sum_x (x - E(X))^2 P_X(x)$$

nth moment:

$$E[X^n] = \sum_x x^n \cdot P_X(x)$$

$$\text{ex: } \text{var}(X) = E[(X - E(X))^2]$$

$$= \sum_x (x - E(X))^2 P_X(x)$$

$$= \sum_{x=-4}^4 \frac{1}{9} \cdot (x^2)$$

$$= \frac{1}{9} (16 + 9 + 4 + 1 + 0 + 1 + 4 + 9 + 16)$$

$$= \frac{60}{9}$$

# Properties of Mean and Variance:

(11)

$$Y = ax + b$$

$$E(Y) = E(ax + b) = \sum_x (ax + b) P_X(x) = a \underbrace{\sum_x x P_X(x)}_{E(X)} + \underbrace{\sum_x b P_X(x)}_b$$

$$E(Y) = a E(X) + b$$

$$V(Y) = \sum_x (ax + b - E(ax + b))^2 P_X(x)$$

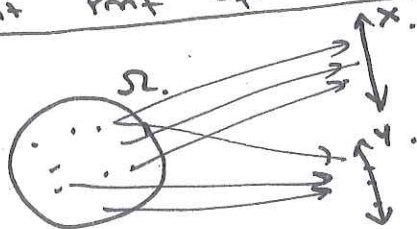
$$= \sum_x (ax + b - a E(X) - b)^2 P_X(x)$$

$$= \sum_x a^2 (x - E(X))^2 P_X(x) = a^2 \underbrace{\sum_x (x - E(X))^2 P_X(x)}_{V(X)}$$

$$V(Y) = a^2 V(X)$$

$$\begin{aligned} V(Y) = E\{(X - E(X))^2\} &= E[X^2 - 2E(X)X + (E(X))^2] \\ &= E(X^2) - 2E(X) \cdot E(X) + E(X)^2 \\ &= E(X^2) - (E(X))^2 \end{aligned}$$

## Joint pmf of Multiple R.V.s:



same experiment.

Let  $X$  and  $Y$  be two discrete r.v.s defined on the sample space  $\Omega$  of the same experiment. Joint pmf

$$\begin{aligned} P_{X,Y}(x,y) &= P(X=x \text{ and } Y=y) \\ &= P(X=x, Y=y) \end{aligned}$$

$$P((X,Y) \in A) = \sum_{(x,y) \in A} P_{X,Y}(x,y)$$

ex:

|   |                |                |                |                      |
|---|----------------|----------------|----------------|----------------------|
|   |                |                |                | $\rightarrow P_X(4)$ |
| 4 | 0              | $\frac{1}{20}$ | $\frac{1}{20}$ | $\frac{1}{20}$       |
| 3 | $\frac{1}{20}$ | $\frac{2}{20}$ | $\frac{3}{20}$ | $\frac{1}{20}$       |
| 2 | $\frac{1}{20}$ | $\frac{2}{20}$ | $\frac{3}{20}$ | $\frac{1}{20}$       |
| 1 | $\frac{1}{20}$ | $\frac{1}{20}$ | $\frac{1}{20}$ | 0                    |
|   | 1              | 2              | 3              | 4                    |
|   |                |                | $\downarrow$   | $P_X(3)$             |

2-D table to show  $P_{X,Y}$ .

$$P_{X,Y}(2,3) = P(X=2, Y=3) = \frac{2}{20}$$

$$P_X(4) = 0 + \frac{1}{20} + \frac{1}{20} + \frac{1}{20}$$

$$P_X(3) = \frac{1}{20} + \frac{3}{20} + \frac{3}{20} + \frac{1}{20}$$

calculate pmf of  $x$  and  $y$ :

$$P_X(x) = \sum_y P_{X,Y}(x,y)$$

$$P_Y(y) = \sum_x P_{X,Y}(x,y)$$

$$E[g(x,y)] = \sum_x \sum_y g(x,y) P_{X,Y}(x,y)$$

$$E[ax+by+c] = aE[x] + bE[y] + c$$

ex: Find pmf of  $z = x+2y$ .

possible values for  $z$ : 3, 4, 5, 6, 7, 8, 9, 10, 11, 12.

$$P_z(3) = \sum_{\{(x,y) | x+2y=3\}} P_{X,Y}(x,y) = P_{X,Y}(1,1) = \frac{1}{20}$$

$$P_z(7) = \sum_{\{(x,y) | x+2y=7\}} P_{X,Y}(x,y) = P_{X,Y}(1,3) + P_{X,Y}(3,2) = \frac{1}{20} + \frac{3}{20} = \frac{4}{20}$$

Note: more than Two Rvs:

$$P_{X,Y,Z}(x,y,z) = P(X=x, Y=y, Z=z)$$

$$P_{X,Y}(x,y) = \sum_z P_{X,Y,Z}(x,y,z)$$

$$P_X(x) = \sum_{y,z} P_{X,Y,Z}(x,y,z)$$

$$E[ax+by+cz+d] = aE[x] + bE[y] + cE[z] + d$$

$$E[a_1x_1 + a_2x_2 + \dots + a_nx_n] = a_1E(x_1) + a_2E(x_2) + \dots + a_nE(x_n)$$

ex:  $x_i$ :  $i$ -th Bernoulli r.v.  $\therefore = \begin{cases} 1, & \text{if success} \\ 0, & \text{if failure} \end{cases}$  (p).

$$y = \sum_{i=1}^n x_i : \# \text{ of successes in } n \text{ trials.}$$

$y$ : Binomial r.v.  $(n, p)$ .

ex: Each day the price of a stock moves up by one point or stays constant with probabilities  $\frac{3}{4}$  and  $\frac{1}{4}$  respectively.

What is the expected gain after six days?

$x_i = \begin{cases} 1, & \text{if stock up by one point} \\ 0, & \text{if stock stays constant.} \end{cases}$

$x_i \sim \text{Bernoulli}(\frac{3}{4})$

$Y = \sum_{i=1}^6 x_i$  : Binomial with  $(6, \frac{3}{4})$

$$E[Y] = E\left(\sum_{i=1}^6 x_i\right) = \sum_{i=1}^6 E(x_i) = \sum_{i=1}^6 \frac{3}{4} = 6 \cdot \frac{3}{4} = \frac{9}{2}$$

$$E(x_i) = 1 \cdot P(x_i=1) + 0 \cdot P(x_i=0) = \frac{3}{4}$$

ex: Suppose  $n$  people throw their hats in a box and then picks one hat at random. What is the expected value of the # of people that get back their own hat?

For  $i$ th person,  $x_i = \begin{cases} 1, & \text{if select his own hat} \\ 0, & \text{otherwise.} \end{cases}$

$$P(x_i=1) = \frac{1}{n}, \quad P(x_i=0) = 1 - \frac{1}{n}$$

$$E(x_i) = 1 \cdot \frac{1}{n} + 0 \cdot \left(1 - \frac{1}{n}\right) = \frac{1}{n}$$

$X$ : # of people that get back their own hat

$$X = x_1 + x_2 + \dots + x_n$$

$$E(X) = E(x_1) + E(x_2) + \dots + E(x_n) = n \cdot \frac{1}{n} = 1$$

### Conditioning:

Remember conditional probability.

$A, B$  two events:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Bayes' rule:

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Total probability theorem:

$\bigcup A_i = \Omega$  , no intersection of  $A_1, A_2, \dots, A_n$ .

$$P(B) = P(B \cap A_1) + P(B \cap A_2) + \dots + P(B \cap A_n) \\ = P(B|A_1) \cdot P(A_1) + P(B|A_2) \cdot P(A_2) + \dots + P(B|A_n) \cdot P(A_n)$$

conditional pmf  $\rightarrow$  conditional expectation

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PMF conditional on event:

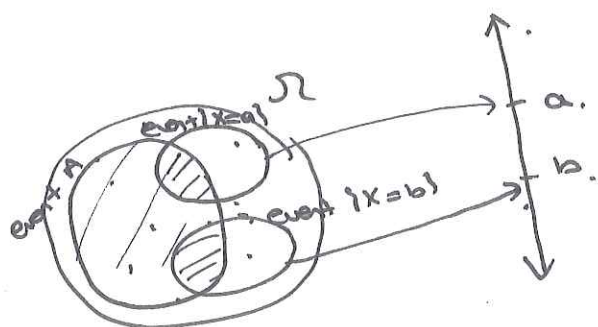
$$P_{X|A}(x) = P(X=x|A) = \frac{P(\{X=x\} \cap A)}{P(A)}$$

$\{X=x\}$  disjoint, union  $= \Omega$ .

$$P(A) = \sum_x P(\{X=x\} \cap A)$$

$$\sum_{x \in \Omega_x} P_{X|A}(x) = \sum_x \frac{P(\{X=x\} \cap A)}{P(A)} = \frac{\sum_x P(\{X=x\} \cap A)}{P(A)} = 1.$$

valid pmf  $\checkmark$ .



ex: Roll a fair six-sided die.  $A = \{\text{roll is even number}\}$

$$P_{X|A} = ?$$

$$P_{X|A}(x) = \frac{P(X=x | \text{roll is even})}{P(\text{roll even})} = \frac{P(X=x, \text{roll even})}{1/2}$$

$$= \begin{cases} 1/3 & , x=2,4,6 \\ 0 & , \text{otherwise} \end{cases}$$

PMF conditional on another r.v.:

$$P_{X|Y}(x|y) = P(X=x|Y=y) = \frac{P(X=x, Y=y)}{P(Y=y)} = \frac{P_{XY}(x,y)}{P_Y(y)}$$

For fixed  $y$ ,  $P_{X|Y}(x|y)$  is pmf.

$$\sum_x P_{X|Y}(x|y) = \sum_x \frac{P_{XY}(x,y)}{P_Y(y)} = \frac{P_Y(y)}{P_Y(y)} = 1.$$

same way:

$$P_{XY}(x,y) = P_{X|Y}(x|y) \cdot P_Y(y)$$

$$E[X|A] = \sum_x x \cdot P_{X|A}(x)$$

$$E[g(X)|A] = \sum_x g(x) \cdot P_{X|A}(x)$$

$$E[X|Y=y] = \sum_x x \cdot P_{X|Y}(x|y)$$

$$* E[X] = \sum_y P_Y(y) E(X|Y=y)$$

Proof:  $P_X(x) = \sum_y P_{X,Y}(x,y) = \sum_y P_Y(y) \cdot P_{X|Y}(x|y)$

$$E[X] = \sum_x x \cdot P_X(x)$$

$$= \sum_x x \cdot \sum_y P_Y(y) \cdot P_{X|Y}(x|y)$$

$$= \sum_y P_Y(y) \underbrace{\sum_x x \cdot P_{X|Y}(x|y)}_{E[X|Y=y]}$$

Similar idea.

$A_1, A_2, \dots, A_n$  - partition of sample space

$$E[X] = \sum_i P(A_i) E[X|A_i]$$

ex: Messages transmitted from Boston to NY with prob 0.5, to Chicago with prob 0.2. The transit time of message  $X$  is random. Its mean is 0.05 sec if destined to NY, 0.1 sec if to Chicago, 0.3 if to SF.  $E[X] = ?$

$$E[X|NY] = 0.05$$

$$E[X|C] = 0.1$$

$$E[X|SF] = 0.3$$

$$NY \cup C \cup SF = \Omega, \text{ no intersection}$$

$$E[X] = P(NY) \cdot E[X|NY] + P(C) \cdot E[X|C] + P(SF) \cdot E[X|SF]$$

$$= 0.5 \times 0.05 + 0.2 \times 0.1 + 0.3 \times 0.3 = 0.115$$

## Independence:

Remember independence of events:

$$P(A|B) = P(A)$$

$$P(B|A) = P(B)$$

$$P(A, B) = P(A) \cdot P(B)$$

1) Independence of a r.v. from an event.

$$P(Z=x, A) = P(Z=x) \cdot P(A) = P_Z(x) \cdot P(A)$$

$$P_{Z|A}(x) = P(Z=x | A) = P(Z=x) = P_Z(x)$$

ex: Two tosses of a fair coin.

$A = \#$  of heads is even

$X = \#$  of heads.

pmf of  $X$ ?

$$P_Z(x) = \begin{cases} 1/4, & x=0 \\ 1/2, & x=1 \\ 1/4, & x=2 \end{cases}$$

pm of  $X$  conditioned on  $A$ ?

$$P_{Z|A}(x) = \frac{P(Z=x, A)}{P(A)} = \begin{cases} 1/2, & x=0 \\ 0, & x=1 \\ 1/2, & x=2 \end{cases}$$

$$P(A) = \frac{1}{2}$$

Since  $P_{Z|A}(x) \neq P_Z(x) \Rightarrow A$  and  $Z$  are not independent.

2) Independence of r.v.s.

$$P(Z=x, Y=y) = P(Z=x) P(Y=y) \text{ for all } x, y.$$

$$P_{Z,Y}(x,y) = P_Z(x) \cdot P_Y(y)$$

$$\text{or } P_{Z,Y}(x|y) = P_Z(x) \text{ for all } y \text{ and } x.$$

$$E(XY) = E(X) \cdot E(Y)$$

If  $X$  and  $Y$  indep

$$\begin{aligned} \rightarrow E(XY) &= \sum_x \sum_y xy P_{Z,Y}(x,y) \\ &= \sum_x \sum_y xy P_Z(x) \cdot P_Y(y) \\ &= \sum_x x \cdot P_Z(x) \sum_y y P_Y(y) \\ &= E(X) \cdot E(Y) \end{aligned}$$

Similarly,

If  $X, Y$  indep,  $E[g(X)h(Y)] = E[g(X)]E[h(Y)]$ .

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If  $X, Y$  indep,  $Var(X+Y) = Var(X) + Var(Y)$

3) Conditional independence

$X$  and  $Y$  conditionally indep given event  $A$  if

$$P_{X,Y|A}(x,y) = P_{X|A}(x) \cdot P_{Y|A}(y) = P_{X|A}(x) P_{Y|A}(y)$$

or  $P_{X,Y|A}(x,y) = P_{X|A}(x)$  for all  $x,y$ .

ex:

|   |                 |                 |                 |                 |
|---|-----------------|-----------------|-----------------|-----------------|
| 4 | <del>1/20</del> | <del>2/20</del> | <del>2/20</del> | 0               |
| 3 | <del>2/20</del> | <del>4/20</del> | <del>1/20</del> | <del>2/20</del> |
| 2 | 0               | 1/20            | 3/20            | 1/20            |
| 1 | 0               | 1/20            | 0               | 0               |
|   | 1               | 2               | 3               | 4               |

$x$

Are  $X$  and  $Y$  indep? No.  $P_{XY}(3,1) = 0 \neq P_X(3) \cdot P_Y(1)$

Given that  $A = \{X \leq 2, Y \geq 3\}$ , are  $X$  and  $Y$  indep?

$$P_{X,Y|A}(1,4) = \frac{P(X=1, Y=4, A)}{P(A)} = \frac{\frac{1}{20}}{\frac{9}{20}} = \frac{1}{9}$$

$$P_{X|A}(1) = \frac{P(X=1, A)}{P(A)} = \frac{\frac{3}{20}}{\frac{9}{20}} = \frac{1}{3}$$

$$P_{Y|A}(4) = \frac{P(Y=4, A)}{P(A)} = \frac{\frac{3}{20}}{\frac{9}{20}} = \frac{1}{3}$$

show for all 4 elements that

$$P_{X,Y|A}(x,y) = P_{X|A}(x) \cdot P_{Y|A}(y)$$

$X$  and  $Y$  indep.

Independence of more than two r.v.s:

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$X_1, X_2, \dots, X_n$  indep if

$$P_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n) = P_{X_1}(x_1) P_{X_2}(x_2) \dots P_{X_n}(x_n)$$

$$Var(X_1 + X_2 + \dots + X_n) = Var(X_1) + Var(X_2) + \dots + Var(X_n)$$

ex:  $X_i$ : Bernoulli:  $(p)$ .

$$X = \sum_{i=1}^n X_i = X_1 + X_2 + \dots + X_n \leftarrow \text{indep.}$$

$$E[X] = np.$$

$$Var(X) = Var\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n Var(X_i) = n \cdot Var(X_i) = np(1-p).$$

Name:

ENGR 200  
QUIZ 3

A fair coin is tossed repeatedly and independently until two consecutive heads or two consecutive tails appear. Find the **pmf** and the **expected value** of the number of tosses.

$$E[X] = \sum_{i=1}^{\infty} i \cdot (1-p)^{i-1} p.$$

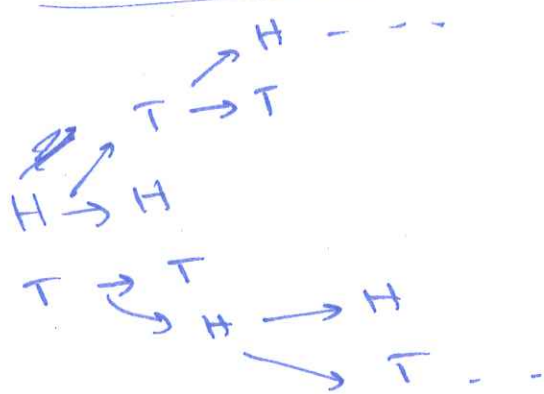
$$= \sum_{i=1}^{\infty} i \cdot q^{i-1} (1-q)$$

$$= \sum_{i=1}^{\infty} i q^{i-1} - i q^i$$

$$= \sum_{i=0}^{\infty} (i+1) q^i - i q^i$$

$$= \sum_{i=0}^{\infty} i q^i + q^i - i q^i$$

$$= \sum_{i=0}^{\infty} q^i = \frac{1}{1-q} = \frac{1}{p}.$$



$X$ : # of tosses until  
possible values 2, 3, ...

$$P(X=2) = \frac{1}{2}$$

$$P(X=3) = \frac{1}{2} \cdot \frac{1}{2}$$

$$P(X=i) = \left(\frac{1}{2}\right)^{i-2} \cdot \frac{1}{2}.$$

$$E[X] = E[X_0] \neq 1.$$

$$= \frac{1}{1/2} + 1 = \underline{\underline{3}}$$

Name:

ENGR 200  
QUIZ 4

Alice passes through four traffic lights on her way to work, and each light is equally likely to be green or red, independent of the others. Find the mean and variance of the number of red lights that Alice encounters. (Explain clearly every step of the derivation.)

$$X_i = \begin{cases} 1, & \text{i-th light red} \\ 0, & \text{otherwise} \end{cases}$$

$$P(X_i = 1) = \frac{1}{2}$$

$$X = \sum_{i=1}^4 X_i$$

$$E(X) = E\left(\sum_{i=1}^4 X_i\right)$$

$$\text{Var}(X) = \sum_{i=1}^4 \text{Var}(X_i)$$

# Quiz #1:

Special chess tournament, play one game with each of three opponents. Win tournament if win two games in a row. Probability of a win against each of three opponents is 0.12, 0.69, 0.77. What is the probability of winning tournament, assume you choose optimal order?

$$\begin{aligned} \{ \text{win tournament} \} &= \{ \overset{\downarrow W}{1^{\text{st}} \text{ win}}, 2^{\text{nd}} \overset{\downarrow W1}{\text{win}}, 3^{\text{rd}} \text{ lose} \} \cup \\ &\quad \{ 1^{\text{st}} \text{ win}, 2^{\text{nd}} \overset{\downarrow W2}{\text{win}}, 3^{\text{rd}} \text{ win} \} \cup \\ &\quad \{ 1^{\text{st}} \text{ lose}, 2^{\text{nd}} \overset{\downarrow W3}{\text{win}}, 3^{\text{rd}} \text{ win} \} \end{aligned}$$

$$\begin{aligned} P(\{ \text{win tournament} \}) &= P(\{W\}) \\ &= P(W_1 \cup W_2 \cup W_3) \\ &= P(W_1) + P(W_2) + P(W_3) \end{aligned}$$

optimal ordering: 0.12, 0.77, 0.69.  
 $P_1 \quad P_2 \quad P_3$

$$\begin{aligned} &= P_1 \cdot P_2 \cdot (1 - P_3) \\ &\quad + P_1 \cdot P_2 \cdot P_3 \\ &\quad + (1 - P_1) \cdot P_2 \cdot P_3 \\ &= P_1 \cdot P_2 + (1 - P_1) \cdot P_2 \cdot P_3 \end{aligned}$$

Bernoulli r.v. with parameter  $p$ :  $P_X(x) = \begin{cases} 1-p, & x=0 \\ p, & x=1 \\ 0, & \text{oth.} \end{cases}$

Binomial r.v.:  $n$  trials: each indep., identical Bernoulli with parameter  $p$ .  
 with parameters  $(n, p)$

Geometric r.v.: trials until success, each indep., identical Bernoulli with parameter  $p$ .  
 with parameter  $p$

Midterm #1, Mon 25, 18<sup>30</sup>, ENG 250. until section ①  
Quiz #1 at 3pm today. 2.6.1

Bernoulli  
with param  $p$  : 
$$p_X(x) = \begin{cases} 1-p, & x=0 \\ p, & x=1 \\ 0, & \text{othw.} \end{cases}$$

Binomial  
with params  $(n, p)$  :  $n$  trials, each indep, identical Bernoulli with param  $p$ .  
count # of successes in  $n$  trials.

Geometric r.v.  
with param  $p$

# of trials until success, each indep, identical Bernoulli with param  $p$ .

Poisson r.v.  
with param  $\lambda$  :

$$p_X(x) = \frac{e^{-\lambda} \lambda^x}{x!} \quad x=0, 1, \dots$$

count # of events OR  
approximate  $\text{bin}(x; n, p)$  where  $n$  large,  
 $p$  small,  
 $np = \lambda$ .

1) Independence of events:

$$P(A|B) = P(A)$$

$$P(A \cap B) = P(A) \cdot P(B)$$

2) Independence of a r.v. from an event:  
for all  $x$

$$P_{X|A}(x) = P_X(x)$$

3) Independence of r.v.s.

$$P_{X|Y}(x|y) = P_X(x) \quad \text{for all } x, y$$

$$P_{XY}(x, y) = P_X(x) \cdot P_Y(y) \quad \text{" "}$$

$$E[X+Y] = E[X] + E[Y]$$

$$E[XY] = E[X] \cdot E[Y]$$

$$Var(X+Y) = Var(X) + Var(Y)$$

always

only if

" " " "

$X, Y$  indep.

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